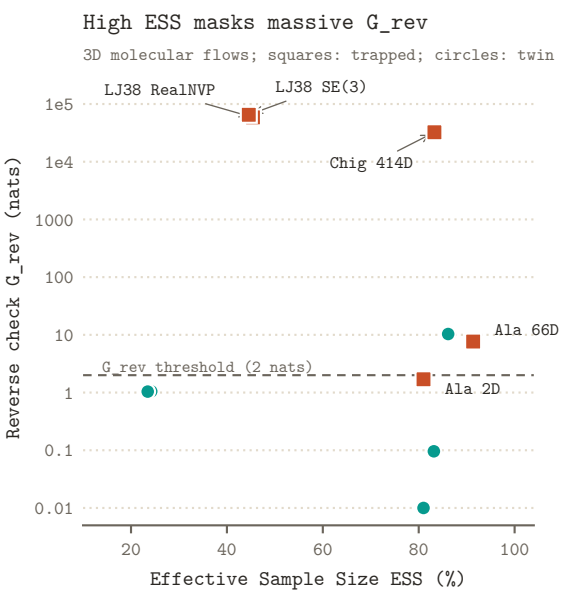
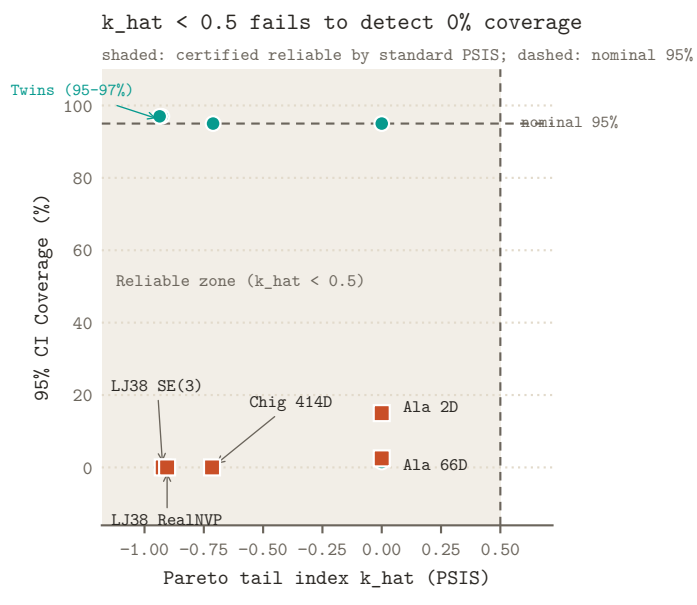
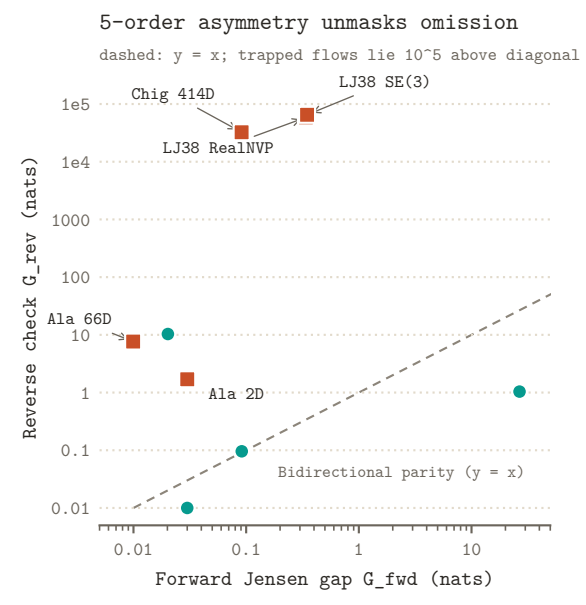

High ESS masks massive G_rev
3D molecular flows; squares: trapped; circles: twin
LJ38 RealNVP
LJ38 SE(3)
Chig 414D
Ala 66D
Ala 2D
G_rev threshold (2 nats)
Reverse check G_rev (nats)
1e5
1e4
1000
100
10
1
0.1
0.01
20
40
60
80
100
Effective Sample Size ESS (%)


k_hat < 0.5 fails to detect 0% coverage
shaded: certified reliable by standard PSIS; dashed: nominal 95%
Twins (95-97%)
nominal 95%
Reliable zone (k_hat < 0.5)
LJ38 SE(3)
Chig 414D
Ala 2D
Ala 66D
LJ38 RealNVP
95% CI Coverage (%)
100
80
60
40
20
0
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
Pareto tail index k_hat (PSIS)


5-order asymmetry unmasks omission
dashed: y = x; trapped flows lie 10^5 above diagonal
Chig 414D
LJ38 SE(3)
LJ38 RealNVP
Ala 66D
Ala 2D
Bidirectional parity (y = x)
Reverse check G_rev (nats)
1e5
1e4
1000
100
10
1
0.1
0.01
0.01
0.1
1
10
Forward Jensen gap G_fwd (nats)